Reflection-Mode Off-Axis Digital Holography System: 3D Surface Reconstruction for Optical Metrology Applications

Project Overview

This project focused on the design, assembly, alignment, and testing of a reflection-mode off-axis digital holography system for optical metrology and 3D surface reconstruction. The system was developed to reconstruct the phase and amplitude information of reflective samples, with potential applications in printed circuit board inspection, semiconductor surface metrology, and non-contact 3D measurement.

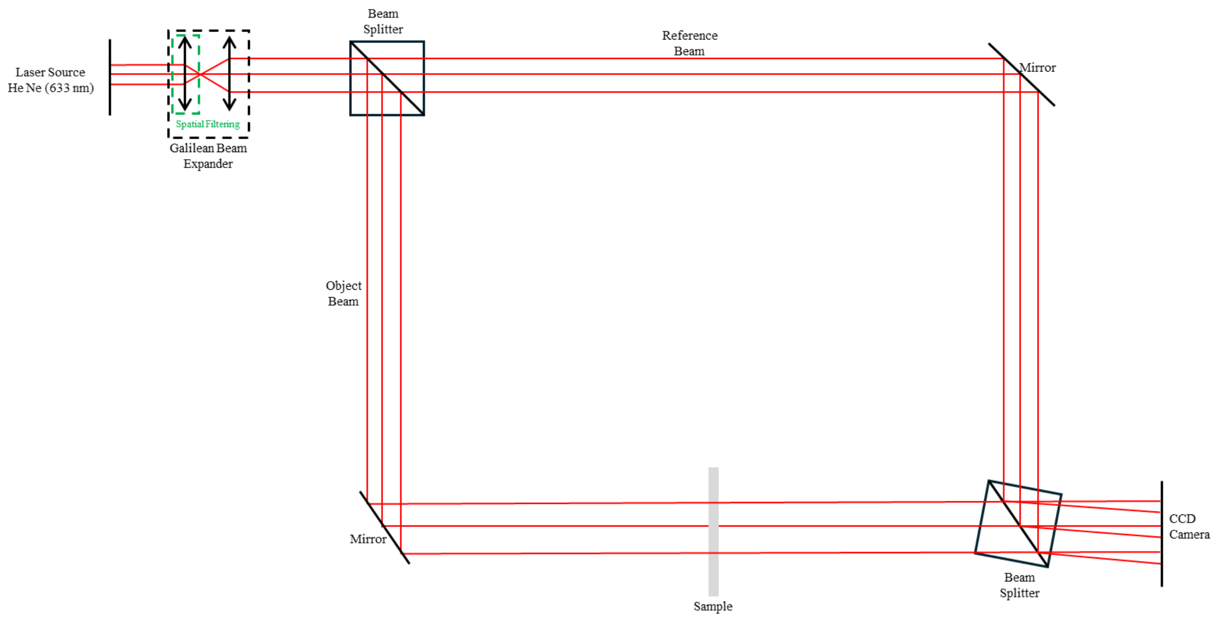

Figure 1. Reflection-mode DH optical system layout.

Unlike transmission-mode digital holography, reflection-mode DH is more suitable for opaque or reflective samples. However, it also introduces additional experimental challenges because the reflected object beam can be weak and highly sensitive to optical noise, unwanted reflections, beam quality, and system alignment. This project addressed these challenges through improved spatial filtering, optical alignment, hologram acquisition, and numerical reconstruction.

Motivation

Optical metrology for industrial and semiconductor applications often requires non-contact measurement of surface profiles, defects, and small topographic variations. Reflection-mode digital holography is attractive because it can reconstruct quantitative phase information from reflective samples without physically contacting the surface.

The main motivation of this project was to develop a reflection-mode DH platform capable of:

- Capturing off-axis holograms from reflective samples
- Reconstructing amplitude and phase information
- Performing numerical propagation for 3D analysis
- Supporting surface metrology of PCB and semiconductor-related samples
- Improving hologram quality through optical filtering and alignment
- Developing a computational workflow for phase reconstruction and visualization

Methodology

The methodology began with the design, assembly, and alignment of a reflection-mode off-axis digital holography system. A coherent laser beam was conditioned using a spatial filtering stage to improve beam quality and reduce unwanted high-order components. After spatial filtering optimization, the object beam was directed toward the reflective sample, while the reference beam was adjusted to interfere with the reflected object beam at a small off-axis angle on the camera sensor.

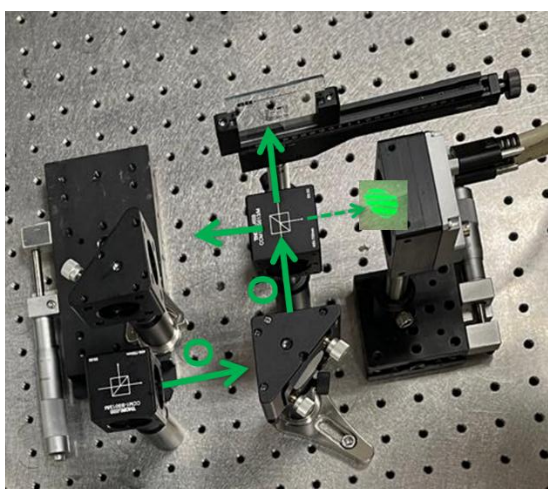

Figure 2. Free space set-up.

The recorded holograms were processed using a Fourier-domain reconstruction workflow. The +1 diffraction order was isolated using spatial filtering, shifted to the center, and transformed back to recover the complex optical field. From this reconstructed field, amplitude and phase images were extracted. Angular spectrum propagation and phase correction were then applied to improve focus, reduce background phase distortion, and evaluate the system's potential for non-contact 3D surface metrology.

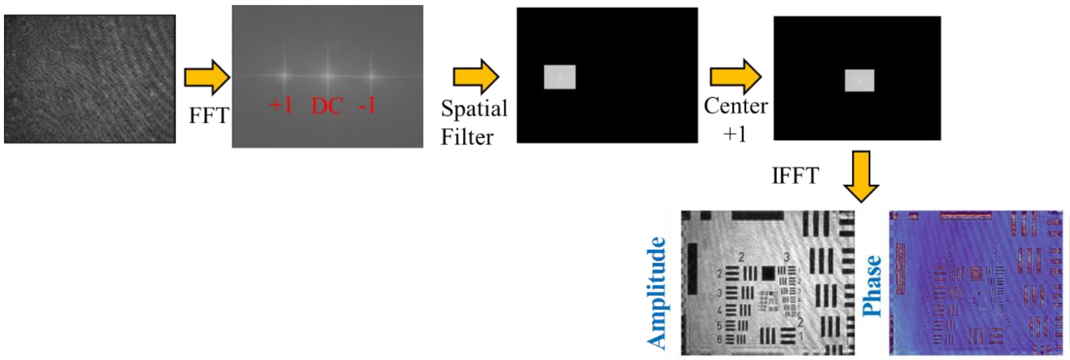

Figure 3. Processing workflow from image acquisition to amplitude and phase.

My Role

My contributions to this project included:

- Designed and assembled the reflection-mode off-axis DHM optical setup.
- Selected and configured optical and optomechanical components for the system.
- Aligned the object and reference beams to obtain stable off-axis interference fringes.
- Developed Python-based routines for hologram reconstruction, spatial filtering, angular spectrum propagation, phase compensation, and result visualization.
- Prepared technical reports and presentations summarizing progress, limitations, and future improvement strategies.

Key Technical Areas

This project demonstrates experience in:

- Reflection-mode digital holography
- Off-axis interferometry
- 3D optical metrology
- PCB and semiconductor surface inspection

Results and Demonstrations

The system was tested through multiple stages of development. Initial holograms showed noise and imperfect beam quality due to spatial filtering limitations. After improving the spatial filtering and alignment, cleaner holograms and clearer Fourier spectra were obtained.

The improved reflection-mode holograms were reconstructed to obtain amplitude and phase images. After numerical propagation and phase correction, the reconstructed phase showed improved quality and better suitability for surface analysis.

Industrial Relevance

Reflection-mode DHM is relevant for non-contact surface metrology of reflective or opaque samples. This makes the approach suitable for applications where transmission imaging is not possible, such as PCB inspection, semiconductor surface analysis, metallic patterns, and reflective microstructures.

Confidentiality Note

Some technical details, raw data, sample information, and implementation details are omitted or generalized due to confidentiality and intellectual property considerations. The project is described here as a reflection-mode digital holography system for optical metrology and 3D reconstruction.

Project Status

This project was developed as part of my R&D internship. After the completion of my internship, the company continued refining and improving this method for future metrology applications involving PCB and semiconductor-related samples.